

Gene

A distinct section of a **chromosome** that is the basic unit of inheritance. Each gene consists of a segment of deoxyribonucleic acid (**DNA**) containing the code that governs the production of a specific **protein**.

Genome

The full set of **genes**, or hereditary information, for a living organism; the human genome consists of 20,000–25,000 genes.

Grey matter

The darker-coloured regions of the brain and spinal cord that comprise mainly **neuron** cell bodies as opposed to their projecting fibres, which form **white matter**.

H**Haemoglobin**

The **protein** in **red blood cells** that combines with oxygen to carry the gas from the lungs and distribute it around the body.

Hair follicle

A pit on the surface of the skin from which hair grows.

Heart valve

One of four structures in the heart that allow the passage of blood in one direction only.

Hepatic

Concerning the **liver**.

Hepatocyte

A type of **liver** cell with many functions, including making **bile**.

Hippocampus

A structure in the **limbic system** in the brain concerned with learning and long-term memory.

Hormone

A chemical released by the **endocrine glands** and some **tissues**. Hormones act on specific receptor sites in other parts of the body.

Hypothalamus

A small structure located at the base of the brain, where the nervous and hormonal systems of the body interact. It is linked to the **thalamus** above and the **pituitary gland** below.

I–K**Ileum**

The final segment of the small intestine, where most absorption of nutrients takes place.

Kidney

One of two bean-shaped organs in the back of the abdominal cavity that filter blood and remove wastes, particularly **urea**.

Killer T-cells

White blood cells that can destroy damaged, infected, or malignant body cells by using proteins called lymphokines.

L**Larynx**

The structure in the neck at the top of the **trachea**, known as the voice box, that contains the **vocal cords**.

Lens

The internal lens of the eye, also called the crystalline lens; it fine-focuses vision by adjusting its curvature. The outer lens is called the **cornea**.

Ligament

A band of **tissue** consisting of **collagen** – a tough, fibrous, elastic **protein**. Ligaments support bones, mainly in and around joints.

Limbic system

A collection of structures in the brain that plays an important role in the automatic (involuntary) body functions, instinctive behaviour, emotions, and the sense of smell.

Lipid

Fatty or oily substance, insoluble in water, with varied roles in the body, including formation of **adipose tissue**, cell membranes (phospholipid), and steroid **hormones**.

Liver

The large **organ** in the upper right abdomen that performs vital chemical functions, including processing of nutrients from the intestines; manufacture of sugars, **proteins**, and fats; detoxification of poisons; and conversion of waste products to **urea**.

Lobe

A rounded projection or subdivision forming part of a larger structure such as the brain, lung, or **liver**.

Lymphatic system

An extensive network of transparent lymph vessels and **lymph nodes**. It returns excess **tissue** fluid to the circulation and combats infections and cancer cells.

Lymph node

A small, oval gland packed with **white blood cells** that acts as a barrier to the spread of infection. Nodes occur in series along lymph vessels.

Lymphocyte

White blood cell that is part of the immune system; it protects against **virus** infections and cancer.

M**Medulla**

The inner part of an **organ**, such as the **kidneys** or adrenal glands. Also refers to the part of the **brainstem** lying immediately above the start of the spinal cord, just in front of the **cerebellum**.

Meninges

Three membrane layers around the brain and spinal cord: the pia mater on the inside, the arachnoid and the dura mater next to the skull.

Meniscus

A crescent-shaped, shock-absorbing pad of **cartilage** found in the knee and some other joints.

Menopause

The end of the reproductive period in women, when the **ovaries** have ceased their production of eggs and menstruation has stopped.

Metabolism

The sum of all the physical and chemical processes that take place in the body.

Middle ear

The air-filled cleft within the temporal bone between the **eardrum** and the outer wall of the inner ear; contains **ossicles**. Also called the tympanic cavity.